Highlights

Trading Intensity, Forecast Disagreement, and Post-Announcement Volume Persistence in European Index Futures

- U.S. macro news generates persistent high-activity periods in European futures.
- Immediate activity predicts 120-minute excess volume in all nine settings.
- Directional concentration adds virtually no explanatory power beyond activity.
- Forecast dispersion attenuates NFP persistence across all three markets.
- NFP produces the largest and longest activity shift in each market.

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Trading Intensity, Forecast Disagreement, and Post-Announcement Volume Persistence in European Index Futures

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ABSTRACT

Trading volume can remain unusually high well after a scheduled macroeconomic announcement. This paper examines event-level variation in that persistence using one-minute FTSE 100, DAX, and CAC 40 futures and U.S. non-farm payroll (NFP), ISM manufacturing, and Consumer Price Index releases from 2000 to 2025. Normal volume is estimated at the same time of day over preceding trading days without the three announcement types. A data-driven turning point for each market and announcement type separates the initial response from a 120-minute persistence interval. Relative to matched non-announcement periods, actual releases generate higher immediate trading activity, larger cumulative positive excess-volume areas, and longer times to the seasonal baseline in all nine market–announcement settings. NFP produces the largest and longest response in each market. Immediate seasonally adjusted activity predicts subsequent excess volume in all nine settings. In the pooled model, adding activity raises adjusted R^2 from 0.130 to 0.395. Directional concentration adds virtually no explanatory power once immediate activity is included. Higher professional forecast dispersion is associated with a smaller cumulative NFP response in all three markets, consistent with a signal-precision channel. The findings characterize post-announcement volume persistence as an unusually high-activity condition associated with the scale of initial trading rather than with concentrated one-sided flow.

1. Introduction

Scheduled macroeconomic announcements often generate a two-stage market response. Prices adjust rapidly when the news becomes public, while trading activity can remain elevated for a considerably longer period. This pattern has been documented in U.S. Treasury, foreign-exchange, and futures markets (Ederington and Lee, 1993; Fleming and Remolona, 1999; Frino and Hill, 2001; Chaboud et al., 2008). The persistence of trading after the initial price response does not necessarily indicate delayed price discovery. Portfolio rebalancing, liquidity transfer, order execution, and continued interpretation of the announcement may all require transactions after the largest price adjustment has occurred. The empirical question examined here is therefore not why trading continues in a structural sense, but what explains event-level variation in the magnitude and persistence of post-announcement trading.

The first question is whether persistence is associated primarily with the intensity of the initial participation shock or with the directional concentration of trading. These features capture different dimensions of the market response. A large increase in total activity may reflect broad participation, portfolio rebalancing, execution splitting, or inventory transfer. A high absolute signed-volume ratio, by contrast, indicates that trading is concentrated more heavily on one side of the market. Raw signed volume increases mechanically with total volume, so directional concentration is measured as absolute tick-rule signed volume divided by total volume. This normalization makes it possible to compare immediate trading activity and directional concentration within the same model. If persistence reflects the scale of market participation, immediate activity should dominate. If one-sided trading pressure is central, normalized directional concentration should retain explanatory power after controlling for activity.

The second question concerns the information environment preceding the announcement. Whereas activity and directional concentration characterize the realized market response, professional forecast dispersion provides an ex ante measure of disagreement about the release. Its relation with subsequent trading is theoretically ambiguous. Differences-of- opinion models imply that heterogeneous beliefs and interpretations can create gains from trade even when average beliefs or prices change little (Harris and Raviv, 1993; Kandel and Pearson, 1995). Under this view, greater forecast dispersion should increase post-announcement trading. An alternative interpretation is that dispersion indicates a

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less precise consensus or a weaker perceived connection between the headline statistic and economic fundamentals. Investors may then place less weight on a given surprise, producing a smaller cumulative response. Consistent with this possibility, Chaboud et al. (2008) document a negative relation between pre-release dispersion and foreign-exchange trading volume, while Born et al. (2023) show that financial markets respond less strongly to news when expectations are more dispersed. Forecast dispersion is therefore treated as a pre-announcement signal-quality condition that may strengthen or attenuate the persistent activity state generated by the release.

The analysis uses individual-contract transaction data for FTSE 100, DAX, and CAC 40 futures and 873 scheduled U.S. macroeconomic announcements: 306 non-farm payroll releases, 307 ISM manufacturing releases, and 260 CPI releases. Continuous futures series are formed using daily contract volume, and transactions are aggregated into one-minute observations. The primary series changes the active contract on the date when the next maturity's daily volume first exceeds that of the current contract. An alternative series implements the change on the following trading day. The three futures markets provide simultaneous responses to a common U.S. information event while representing major European equity indices traded on different exchanges.

Three features of the empirical design separate announcement-induced persistence from ordinary intraday trading patterns. First, normal volume is estimated separately for each time of day using the median log volume over the preceding 20 trading days without NFP, CPI, or ISM releases. This adjustment prevents predictable intraday seasonality from being classified as an announcement response. Second, a data-driven turning point in the average seasonally adjusted volume path separates the initial activity spike from the subsequent digestion phase. The primary persistence outcome is the cumulative positive abnormal volume over a 120-minute interval beginning at that turning point.Third, every announcement is matched to a prior non-announcement trading day at the same time of day. The placebo comparison shows whether the announcement produces an unusually large activity state rather than merely inheriting normal short-horizon volume clustering.

The evidence yields four main findings. First, macroeconomic announcements move all three markets into unusually active trading states. Relative to matched non-announcement periods, the announcements generate higher immediate trading activity, larger cumulative excess-volume areas, and longer times to the seasonal baseline in all nine market–announcement combinations. The magnitude of the state shift differs across announcement types. NFP produces the largest immediate activity increase and the longest-lasting response in each market. Following an actual NFP announcement, the seasonal volume baseline is reached approximately 30 to 34 minutes later than during the corresponding matched non-announcement period.

Second, immediate seasonally abnormal trading activity is a strong predictor of the subsequent cumulative response. The activity coefficient is positive and statistically significant in every market–announcement combination. In the pooled model, which combines the nine market–announcement combinations and clusters standard errors by announcement date, the coefficient on standardized immediate activity is 0.357 ($p < 0.001$). Adding immediate activity raises adjusted R^2 from 0.130 to 0.395, an increase of 26.5 percentage points. The result remains positive across alternative seasonal benchmarks, fixed timing intervals, year effects, winsorization, alternative contract-activation rules, and block-bootstrap quantile regressions.

Third, directional concentration provides virtually no incremental explanatory power. Its pooled coefficient is statistically insignificant, and adding it to the activity model raises adjusted R^2 by only about 0.06 percentage points. Its sign and significance also vary across markets and between the same-day and next-day contract-activation rules. The evidence therefore supports an activity-state explanation rather than a one-sided directional-flow explanation of post-announcement volume persistence.

Fourth, forecast dispersion attenuates the cumulative NFP response. Higher NFP dispersion is associated with a smaller cumulative excess-volume area in FTSE, DAX, and CAC futures. The negative relation remains under alternative seasonal benchmarks and fixed timing intervals and is most stable in FTSE and DAX. No comparable common cross-market pattern appears for ISM or CPI. The NFP result is consistent with a signal-precision interpretation: when professional forecasts are more dispersed, the consensus provides a less precise benchmark for evaluating the release, and the subsequent cumulative trading response is weaker.

The paper makes four contributions. First, it shifts attention from the existence of announcement-related volume spikes to event-level variation in the trading activity that persists after the initial response. Second, it implements a reproducible segmentation of the seasonally adjusted volume path and measures persistence as cumulative positive abnormal volume over a fixed 120-minute interval beginning at the estimated turning point. Third, it distinguishes the intensity of market participation from the directional concentration of trading and shows that participation intensity is the dominant predictor of persistence. Fourth, it links pre-announcement forecast dispersion to the cumulative

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NFP response and provides evidence consistent with a signal-precision channel. Together, the findings characterize post-announcement volume persistence as a high-activity state whose magnitude depends primarily on the initial participation shock and, for NFP, on the precision of the pre-release consensus.

The remainder of the paper is organized as follows. Section 2 reviews the related literature and develops the empirical predictions. Section 3 describes the futures and announcement data and the continuous-contract construction. Section 4 defines the seasonality adjustment, the activity and persistence measures, the empirical specifications, and the matched non-announcement design. Section 5 presents the main results. Section 6 reports the robustness analyses. Section 7 discusses the economic interpretation and limitations, and Section 8 concludes.

2. Related Literature and Hypotheses

2.1. Two-stage adjustment and persistent trading

Scheduled macroeconomic announcements provide a useful empirical setting because their release times are known in advance and their surprises can be measured relative to professional forecasts. Early high-frequency studies show that prices and volatility respond sharply to macroeconomic news (Ederington and Lee, 1993; Andersen et al., 2003). Extending this work to trading activity and liquidity, Fleming and Remolona (1999) document an initial price adjustment followed by a more prolonged period of elevated trading, volatility, and wider spreads in the U.S. Treasury market. Frino and Hill (2001) show that price volatility, trading volume, and quoted spreads also adjust rapidly around scheduled futures-market announcements, although their adjustment paths are not identical. Green (2004) finds that trading becomes more informative after economic news in the Treasury market, while Pasquariello and Vega (2007) relate the price impact of order flow to heterogeneous beliefs and information conditions.

These findings motivate a distinction between the initial market response and the subsequent completion of announcement-related trading. Investors who observe the same headline release may nevertheless differ in their portfolio exposures, risk constraints, interpretations of the report's components, and desired execution speed. Large orders may be divided across time, and liquidity providers may need to absorb and redistribute the resulting inventory. The trading process initiated by the announcement can therefore continue after the largest initial response. The analysis refers to this later period as the digestion phase and measures its cumulative magnitude and duration.

Two related but distinct predictions follow. First, scheduled announcements should move the market into a higher-activity state than an otherwise comparable period at the same time of day. This is a prediction about the level and duration of the activity response generated by the announcement. Second, among announcement events, greater immediate seasonally adjusted activity should predict a larger subsequent cumulative excess-volume response. Intensive trading during the initial phase is consistent with more extensive repositioning, order execution, and inventory transfer. If these processes unfold over time, the scale of the initial activity response should be associated with the amount of abnormal trading that remains after the initial phase.

The second prediction does not require the activity–persistence relation to be unique to announcement windows. Short-horizon dependence in trading activity may also produce a positive relation during matched non-announcement periods. The distinction is therefore between whether announcements generate unusually elevated and prolonged activity states and whether cross-event variation in initial activity predicts the subsequent cumulative response. The matched non-announcement design addresses the first question, while the event-level persistence regressions address the second. Comparisons with matched-period slopes provide additional evidence on whether the activity–persistence relation is itself stronger during announcement windows.

H1 (Announcement-induced activity state). Relative to matched non-announcement periods at the same time of day, macroeconomic announcements generate higher immediate seasonally adjusted trading activity, larger cumulative excess-volume areas, and longer times to the seasonal baseline.

H2 (Activity propagation). Among announcement events, greater immediate seasonally adjusted trading activity predicts a larger subsequent cumulative excess-volume response.

2.2. Directional concentration as a competing explanation

The size of the initial participation shock does not describe whether transactions are balanced across directions or concentrated predominantly on one side of the market. Models of informed and strategic trading link signed order flow to information and price formation (Kyle, 1985). Public announcements may also generate heterogeneous

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interpretations, additional information production, and strategic repositioning after the release (Kim and Verrecchia, 1994; Green, 2004; Pasquariello and Vega, 2007).

Although this literature often focuses on price impact, it also motivates a competing explanation for persistent trading. If the announcement produces concentrated buying or selling pressure, liquidity providers may need additional time to absorb the flow and redistribute positions. Under this explanation, persistence depends not only on how much trading occurs initially but also on how strongly that trading is concentrated in one direction.

Raw absolute signed volume is mechanically related to total trading volume. A larger activity shock can therefore produce a larger signed-volume measure even when the directional composition of trading is unchanged. To separate directionality from activity, I divide absolute tick-rule signed volume by total volume during the initial phase. The resulting measure captures directional concentration rather than the scale of trading. Including immediate activity and directional concentration in the same regression tests whether one-sidedness contains information about the subsequent cumulative response beyond the total participation shock.

H3 (Directional-flow channel). Conditional on immediate trading activity, greater absolute directional concentration during the initial phase predicts a larger subsequent cumulative excess-volume response.

2.3. Forecast disagreement and signal precision

Immediate activity and directional concentration describe how the market responds after the announcement. Forecast dispersion captures a different dimension: the information environment in which that response begins. A wide distribution of professional forecasts may indicate substantial disagreement about the release, but it may also indicate that the consensus provides an imprecise benchmark for evaluating the announced value.

Public information can generate trade because agents interpret a common signal differently (Harris and Raviv, 1993; Kandel and Pearson, 1995). In Kim and Verrecchia (1991), price reactions reflect changes in aggregate beliefs while volume also reflects belief revisions across agents. Bollerslev et al. (2018) formalize this intuition through the volume–volatility elasticity around public news and show that it varies with disagreement.

The relation between ex ante forecast dispersion and post-release volume is nonetheless ambiguous. If dispersion captures heterogeneous priors or differential interpretations, it creates additional motives to trade and should increase both the immediate response and the volume required to complete the post-release adjustment. This prediction follows the differences-of-opinion mechanism in which public information does not necessarily produce belief convergence.

A competing interpretation treats dispersion as an observable signal of forecast quality. Wide disagreement among professional forecasters may indicate that the announcement is difficult to predict, has a noisy relation with underlying fundamentals, or is surrounded by greater model uncertainty. Investors may then place less weight on a given headline surprise, require a higher threshold before changing positions, or adopt a wait-and-see response. Chaboud et al. (2008) report that macro-news volume can be negatively related to ex ante dispersion, while Born et al. (2023) interpret dispersion as a signal about the strength of the news–fundamentals link. The balance of these two mechanisms may differ across announcement types because professional forecasts do not represent the same information structure for labour-market, manufacturing, and price statistics.

The analysis therefore considers competing, announcement-general predictions rather than assuming ex ante that dispersion must have the same sign for NFP, ISM, and CPI.

H4a (Differences-of-opinion channel). Conditional on surprise magnitude and immediate trading activity, greater pre-announcement forecast dispersion predicts a larger cumulative excess-volume response.

H4b (Signal-precision channel). Conditional on surprise magnitude and immediate trading activity, greater pre-announcement forecast dispersion predicts a smaller cumulative excess-volume response when disagreement indicates a less precise consensus signal.

3. Data and Continuous-Futures Construction

3.1. Futures and announcement data

The futures data consist of transaction records for individual FTSE 100, DAX, and CAC 40 futures maturities. Each record contains the transaction date, time, price, and reported volume. FTSE and DAX futures follow quarterly maturity cycles, while CAC futures have monthly maturities. The available histories begin in 1996 for FTSE and DAX and in 1999 for CAC and end on 3 July 2025. The announcement sample begins in 2000, when the futures histories and professional forecast data overlap.

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FTSE, DAX, and CAC futures provide a common international setting for the analysis. They represent major European equity indices but trade on different exchanges and have different underlying constituents and maturity cycles. Each U.S. announcement reaches the three markets at the same instant. Cross-market differences in the measured response therefore reflect reactions to the same information release. The design supports two forms of inference: separate market regressions allow the response to differ across FTSE, DAX, and CAC futures, while pooled models summarize their common response and test cross-market coefficient equality. Because the three observations associated with the same release share a common information event, pooled standard errors are clustered by announcement date.

The announcement data cover U.S. non-farm payrolls, ISM manufacturing, and CPI. The three releases differ in their economic content and timing. NFP provides a broad and closely followed signal of labor-market conditions and aggregate activity. ISM manufacturing summarizes purchasing managers' assessments and is released later in the European trading day. CPI provides information about price pressures and the expected monetary-policy path. These differences may affect both the size and persistence of the trading response. The empirical models therefore allow all coefficients to vary across announcement types rather than imposing a common response to NFP, ISM, and CPI.

The macroeconomic data contain announcement dates and times, released values, Bloomberg consensus forecasts, and the cross-sectional dispersion of professional forecasts. The sample comprises 306 NFP, 307 ISM manufacturing, and 260 CPI announcements. U.S. release times are converted to local exchange times using historical time-zone rules, including the weeks in which U.S. and European daylight-saving schedules do not coincide.

For announcement i of type k, the standardized surprise is

$$\widehat{S}_{i,k} = \frac{A_{i,k} - F_{i,k}}{\mathrm{sd}(A_k - F_k)}, \tag{1}$$

where $A_{i,k}$ is the released value and $F_{i,k}$ is the median professional forecast. Forecast dispersion is standardized separately within each announcement type. The main volume specifications use the absolute standardized surprise, $|\widehat{S}_{i,k}|$, so that equally large positive and negative surprises have the same baseline effect. This restriction is evaluated by adding a negative-surprise indicator and an interaction between that indicator and surprise magnitude. The signed surprise is retained for these tests.

Records with invalid dates or economically implausible prices are removed. One-minute open, high, low, and close prices use all valid price observations. Volume, transaction counts, and directional measures use only records with positive reported volume. Missing reported volume contributes zero to the daily contract-volume totals used to determine roll dates.

3.2. Continuous contracts

I construct a continuous futures series for each market using the same daily-volume rule. For every trading date, reported transaction volume is summed separately across available contract maturities. During the final 30 calendar days before expiration, the series changes to the next maturity when its daily volume first exceeds that of the current contract.

Under the primary convention, the next maturity is used for the entire crossover date. The alternative convention implements the change on the following trading day and therefore avoids using end-of-day volume information to select the contract traded earlier on the crossover date. If no volume crossover occurs before expiration, the series changes contract at expiration. Contract selection is monotonic, so the series cannot return to an earlier maturity after a change has occurred.

Under the primary convention, the continuous series contain 5.82 million FTSE, 6.06 million DAX, and 4.78 million CAC one-minute observations. Validation checks find no duplicate or nonchronological timestamps, OHLC inconsistencies, negative volume, or trading dates assigned to more than one active contract. Indicators for contract changes and the distance to the nearest change date are retained for the rollover robustness tests.

One-minute OHLC prices use all valid price observations. Directional trading is measured only from records with positive reported volume. Trade direction is defined by the sign of the price change relative to the preceding positive-volume transaction. When the price is unchanged, the most recent nonzero direction within the same trading date is carried forward.

Let q_j denote transaction volume and let $d_j \in \{-1, 0, 1\}$ denote tick-rule direction. Directional concentration during the initial response is

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$$D_{i,m,k} = \frac{\left|\sum_{j \in \mathcal{I}_{i,m,k}} d_j q_j\right|}{\sum_{j \in \mathcal{I}_{i,m,k}} q_j}, \tag{2}$$

where $\mathcal{I}_{i,m,k}$ denotes the initial response interval for event i, market m, and announcement type k. The ratio is bounded between zero and one. Higher values indicate that initial trading is more heavily concentrated in one direction, independently of the total volume traded. Because a uniform quote history is unavailable, the measure relies on transaction-price changes rather than quote-based trade classification. It is interpreted as directional concentration, not as a direct measure of trader information or motive.

4. Measuring Transient and Persistent Volume

4.1. Seasonality adjustment and persistence measures

Intraday trading volume follows a pronounced time-of-day pattern, with systematic increases around market openings, closings, and the overlap between European and U.S. trading hours (Admati and Pfleiderer, 1988; Andersen and Bollerslev, 1997). Measuring the announcement response relative to raw volume would therefore confound the effect of the release with normal variation over the trading day.

Let i index announcements, m markets, k announcement types, and r the number of minutes elapsed since the release. For each announcement and minute, the seasonal benchmark is estimated from the same local time of day on the 20 most recent preceding trading days with valid observations. Trading days containing an NFP, CPI, or ISM release are excluded so that announcement-related volume does not enter the benchmark. Only dates preceding the event are used, preventing future information from affecting the seasonal adjustment.

The benchmark log volume is

$$B_{i,m,r} = \text{median}_{d \in \mathcal{D}_{i,20}} \left\{ \log\left(V_{d,m,\tau_i+r} + 1\right) \right\}, \tag{3}$$

where τ_i is the local release time and $\mathcal{D}_{i,20}$ denotes the set of preceding comparison dates. The median is used to reduce the influence of unusually active trading days. The addition of one allows the logarithm to be defined when no volume is reported during a particular minute.

The corresponding benchmark volume in levels is

$$\widetilde{V}_{i,m,r} = \exp(B_{i,m,r}) - 1. \tag{4}$$

Thus, $\widetilde{V}_{i,m,r}$ represents the typical volume observed at the same time of day on recent non-announcement dates. Seasonally adjusted log volume and the volume ratio are

$$\text{SA}_{i,m,r} = \log\left(V_{i,m,\tau_i+r} + 1\right) - B_{i,m,r}, \tag{5}$$

$$R_{i,m,r} = \frac{V_{i,m,\tau_i+r} + 1}{\widetilde{V}_{i,m,r} + 1} = \frac{V_{i,m,\tau_i+r} + 1}{\exp(B_{i,m,r})}. \tag{6}$$

The two measures contain the same information because $\text{SA}_{i,m,r} = \log(R_{i,m,r})$. A value of $\text{SA}_{i,m,r} = 0$, or equivalently $R_{i,m,r} = 1$, indicates volume equal to its time-of-day benchmark. Positive values indicate unusually high volume, while negative values indicate volume below the benchmark. Replacing the 20-day benchmark with the preceding 40 non-announcement trading days provides a robustness check.

The announcement response typically contains a sharp initial increase followed by a more gradual adjustment. Rather than imposing the same initial interval on every market and announcement type, I estimate a turning point, denoted $T^*_{m,k}$, from the average seasonally adjusted volume-ratio path over minutes 0 through 15 after the release. Both the time axis and the average volume-ratio axis are scaled to the unit interval. This normalization prevents the units or range of either axis from mechanically dominating the geometric distance calculation.

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Table 1
Sample Coverage and Estimated Turning Points

Market	Announcement	Available releases	Usable events	T^*
FTSE	CPI	260	257	2
FTSE	ISM	307	287	7
FTSE	NFP	306	297	3
DAX	CPI	260	257	2
DAX	ISM	307	281	2
DAX	NFP	306	298	3
CAC	CPI	260	253	13
CAC	ISM	307	257	4
CAC	NFP	306	274	6

Usable events require sufficient observations before and after the release and at least ten preceding comparison dates for the seasonal benchmark. T^* is the turning point estimated from the average seasonally adjusted volume-ratio path for each market and announcement type under the primary contract-activation rule.

Let p_r denote the scaled point corresponding to minute r. The estimated turning point is

$$T^*_{m,k} = \max\left\{2,\ \arg\max_{r\in\{1,\dots,14\}} \operatorname{dist}\left(p_r, \overline{p_0 p_{15}}\right)\right\}, \tag{7}$$

where $\operatorname{dist}(p_r, \overline{p_0 p_{15}})$ is the perpendicular distance between p_r and the line joining the first and last points of the path. The point with the greatest distance identifies the strongest change in the slope of the average response. The lower bound ensures that the initial phase contains at least two one-minute observations. This geometric turning-point rule adapts the elbow heuristic of Thorndike (1953); it is used as a reproducible segmentation procedure rather than as a structural-break estimator.

The boundary is estimated separately for each market–announcement combination, not for each individual event. An event-specific boundary would allow an event's own volume path to determine the interval over which its activity and persistence are measured. Using a common boundary within each market–announcement combination prevents this mechanical dependence and makes event outcomes comparable, while still allowing the average response profile to differ across markets and announcement types. The estimated boundaries are reported in Table 1.

Two robustness checks reduce dependence on the estimated boundary. First, a fixed specification defines the first five minutes after the release as the initial phase and the following 120 minutes as the persistence interval. Second, the seasonal benchmark is expanded from 20 to 40 preceding non-announcement trading days. Both alternatives produce the same main conclusions.

Immediate seasonally adjusted trading activity is measured over the initial phase, $r = 0, \dots, T^*_{m,k} - 1$. It is defined as

$$Activity_{i,m,k} = \log\left[\frac{1 + \sum_{r=0}^{T^*_{m,k}-1} V_{i,m,r}}{1 + \sum_{r=0}^{T^*_{m,k}-1} \widetilde{V}_{i,m,r}}\right]. \tag{8}$$

The numerator is the total observed volume during the initial phase, while the denominator is the total volume implied by the time-of-day benchmark over the same minutes. A positive value therefore indicates that the announcement generated more initial trading than would normally be expected at that time of day. The logarithmic transformation reduces right-skewness and allows the measure to be interpreted as a proportional activity shock.

The primary persistence measure is the cumulative positive seasonally adjusted log volume during the 120 minutes beginning at the estimated boundary:

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$$\mathrm{AUC}^{120}_{i,m,k} = \sum_{r=T^*_{m,k}}^{T^*_{m,k}+119} \max\left(\mathrm{SA}_{i,m,r}, 0\right). \tag{9}$$

The measure is nonnegative and unbounded; its magnitude reflects both the intensity and duration of trading above the seasonal benchmark. Only positive seasonal deviations enter the measure. Minutes with volume below the time-of-day benchmark are set to zero rather than allowed to offset minutes with unusually high volume. This treatment follows the economic object of interest: the total amount of elevated trading that remains after the initial response. A net sum of positive and negative deviations would instead allow a quiet minute to erase an earlier period of unusually intense trading, even though that trading had already occurred.

The 120-minute interval balances two considerations. It is long enough to capture the persistent right tail of the announcement response, particularly following NFP releases, whose median time to the seasonal baseline is approximately 54–57 minutes across the three markets. It also reduces right-censoring relative to a one-hour interval while remaining short enough to focus on trading plausibly connected to the announcement. The same horizon is applied to every event, preserving comparability across markets and announcement types. As a shorter-horizon robustness check, I also measure persistence from $T^*_{m,k}$ through $T^*_{m,k} + 60$, inclusive, which contains 61 timestamped one-minute observations.

The dependent variable in all event-level persistence regressions is $Y_{i,m,k} = \log(1 + \mathrm{AUC}^{120}_{i,m,k})$. This transformation retains zero values and limits the influence of exceptionally large cumulative responses. I estimate a sequence of models that separates announcement characteristics, the scale of the immediate activity response, and directional concentration.

Two secondary measures describe the shape and duration of the response. First, a baseline-anchored autoregression is estimated without an intercept:

$$\mathrm{SA}_{i,m,r} = \rho_{i,m,k}\mathrm{SA}_{i,m,r-1} + u_{i,m,r}, \qquad r = T^*_{m,k} + 1, \ldots, T^*_{m,k} + 119. \tag{10}$$

Because seasonally adjusted log volume is centered at zero when trading returns to its time-of-day benchmark, omitting the intercept anchors the process to that benchmark. When $0 < \rho_{i,m,k} < 1$, the associated half-life is

$$HL_{i,m,k} = -\frac{\log(2)}{\log(\rho_{i,m,k})}. \tag{11}$$

The half-life provides a compact decay summary but is treated as secondary because actual volume paths need not decline smoothly.

Second, time to baseline is the number of minutes from $T^*_{m,k}$ until the volume ratio falls to 1.25 or below for five consecutive minutes. Requiring five consecutive observations prevents a single low-volume minute from being classified as the end of the response. The threshold allows volume to remain up to 25% above its seasonal benchmark. Events that do not satisfy the criterion within the 120-minute persistence interval are right-censored at 120 minutes.

4.2. Empirical models and inference

The empirical analysis follows a nested model sequence aligned with the hypotheses developed in Section 2. In all specifications, the dependent variable is $Y_{i,m,k} = \log(1 + \mathrm{AUC}^{120}_{i,m,k})$.

The first specification relates this event-level persistence outcome to announcement characteristics and the pre-release market state:

$$Y_{i,m,k} = \alpha_{m,k} + \beta_1|\widehat{S}|_{i,k} + \beta_2 Disp_{i,k} + \beta_3 GK_{i,m,k} + \varepsilon_{i,m,k}. \tag{12}$$

Here, $|\widehat{S}|_{i,k} = |\widehat{S}_{i,k}|$ is the absolute standardized surprise, $Disp_{i,k}$ is standardized professional forecast dispersion, and $GK_{i,m,k}$ is Garman–Klass volatility measured during the 15 minutes preceding the release (Garman and Klass, 1980). This fist specification provides a benchmark for the explanatory power of announcement magnitude, the pre-announcement information environment, and prevailing price variation.

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The second specification adds immediate seasonally adjusted trading activity:

$$Y_{i,m,k} = \alpha_{m,k} + \beta_1 |\widehat{S}|_{i,k} + \beta_2 Disp_{i,k} + \beta_3 GK_{i,m,k} + \beta_4 Activity_{i,m,k} + \varepsilon_{i,m,k}. \tag{13}$$

The coefficient β_4 tests the activity-propagation prediction in H2. Comparing this specification with the previous one also shows how much incremental explanatory power is provided by the size of the initial participation shock. Because activity is standardized within each market–announcement combination, β_4 measures the change in log cumulative excess volume associated with a one-standard-deviation increase in immediate activity.

The third specification adds directional concentration:

$$Y_{i,m,k} = \alpha_{m,k} + \beta_1 |\widehat{S}|_{i,k} + \beta_2 Disp_{i,k} + \beta_3 GK_{i,m,k} + \beta_4 Activity_{i,m,k} + \beta_5 D_{i,m,k} + \varepsilon_{i,m,k}. \tag{14}$$

The coefficient β_5 tests H3. It indicates whether the directional concentration of initial trading predicts subsequent persistence after controlling for the total participation shock. Comparing this third specification with the second provides a direct assessment of whether directionality adds explanatory power beyond activity. The dispersion coefficient in M2 and M3 tests the competing predictions in H4a and H4b conditional on the realized initial activity response.

These three specifications are first estimated separately for each of the nine market–announcement combinations. This approach allows all coefficients to vary freely across FTSE, DAX, and CAC futures and across NFP, ISM, and CPI releases. Market-specific regressions use Newey–West standard errors with lag length $\lfloor N_{m,k}^{1/3} \rfloor$ (Newey and West, 1987). Activity, directional concentration, and Garman–Klass volatility are standardized within each market–announcement combination. Surprise and forecast dispersion are standardized within announcement type.

The separate regressions reveal heterogeneity but do not provide a single estimate of the average relation across European markets. They also do not account directly for the fact that FTSE, DAX, and CAC responses to the same U.S. release share a common information event. I therefore stack the nine market–announcement combinations and estimate

$$Y_{i,m,k} = \alpha_{m,k} + \beta_1 |\widehat{S}|_{i,k} + \beta_2 Disp_{i,k} + \beta_3 GK_{i,m,k} + \beta_4 Activity_{i,m,k} + \beta_5 D_{i,m,k} + \varepsilon_{i,m,k}, \tag{15}$$

where $\alpha_{m,k}$ denotes market–announcement fixed effects. These fixed effects absorb persistent differences in average volume persistence across the nine combinations. Because the continuous regressors are standardized within combination, the pooled coefficients summarize average within-combination associations.

Standard errors are clustered by announcement date. This permits residuals from FTSE, DAX, and CAC futures generated by the same U.S. release to be correlated. To examine whether the activity relation differs across combinations, I also estimate a fully interacted specification in which immediate activity is interacted with the nine market–announcement indicators. The resulting coefficients reproduce combination-specific activity slopes within a common model and permit a joint test of whether the nine slopes are equal.

A further pooled extension examines whether global uncertainty changes either the level of persistence or the propagation of the initial activity shock. Let VIX_{i-1} denote the standardized logarithm of the VIX closing level on the most recent U.S. trading day strictly before announcement i. The model is

$$Y_{i,m,k} = X'_{i,m,k}\beta + \gamma_1 VIX_{i-1} + \gamma_2 \left(Activity_{i,m,k} \times VIX_{i-1}\right) + \alpha_{m,k} + \varepsilon_{i,m,k}, \tag{16}$$

where $X_{i,m,k}$ contains absolute surprise, forecast dispersion, Garman–Klass volatility, immediate activity, and directional concentration. The coefficient γ_1 captures a shift in average persistence associated with the prior global uncertainty state. The interaction coefficient γ_2 tests whether the same initial activity shock produces a different cumulative response when global uncertainty is high. Using the most recent strictly prior VIX close prevents announcement-day price movements from entering the state variable.

Additional specifications examine asymmetry and interactions between announcement characteristics. Surprise-sign asymmetry is tested by adding an indicator for negative surprises and its interaction with absolute surprise

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magnitude. A separate model adds an interaction between absolute surprise and forecast dispersion to test whether the response to a given surprise varies with the precision of the pre-announcement consensus.

Each interaction is estimated separately across the nine market–announcement combinations. Testing several related coefficients increases the probability of obtaining an apparently significant result by chance. For each set of nine related tests, I therefore report both Benjamini–Hochberg and Holm adjusted p-values (Benjamini and Hochberg, 1995; Holm, 1979). The Benjamini–Hochberg procedure controls the expected proportion of false rejections among the rejected hypotheses, while the more conservative Holm procedure controls the probability of at least one false rejection within the set. The main forecast-disagreement conclusions are based on the additive dispersion coefficients; the interaction models are treated as supplementary mechanism tests.

The initial seasonally adjusted volume response is analyzed separately. Because immediate activity and directional concentration are calculated from the same initial interval, using them as regressors for the initial volume response would condition on contemporaneous components of the outcome. The initial response is therefore estimated using the first specification, which contains only announcement characteristics and pre-release Garman–Klass volatility.

The baseline-anchored log half-life is used as a secondary persistence outcome in the fist three models. In addition, quantile regressions at $\tau \in \{0.25, 0.50, 0.75, 0.90\}$ examine whether the activity relation is concentrated in a particular part of the persistence distribution. Inference for the quantile estimates uses a moving-block bootstrap to preserve temporal dependence across nearby announcements.

The matched non-announcement analysis evaluates whether the level and propagation of activity observed after actual releases differ from ordinary short-horizon volume clustering. Each announcement is paired with a nearby prior trading date without an NFP, CPI, or ISM release, using the same local time of day. Actual and matched periods receive the same seasonal adjustment and use common fixed intervals: the first five minutes for initial activity and the following 120 minutes for persistence. Fixed intervals ensure that the actual and matched observations are measured identically.

The first comparison uses paired tests of differences in immediate activity, $\log(1 + \mathrm{AUC}^{120})$, and time to baseline. To compare activity-propagation slopes, actual and matched observations are then stacked within each market–announcement combination:

$$Y_j = \delta_0 + \delta_1 Activity_j + \delta_2 Actual_j + \delta_3 \left(Activity_j \times Actual_j \right) + Z_j'\theta + e_j, \tag{17}$$

where $Actual_j$ equals one for an announcement observation and zero for its matched non-announcement observation. The vector Z_j contains pre-release Garman–Klass volatility and directional concentration. The coefficient δ_1 is the activity slope during matched non-announcement periods, while δ_3 measures the additional activity propagation associated with an actual announcement. A positive δ_3 therefore indicates that immediate activity predicts subsequent persistence more strongly after the release than during an otherwise comparable period at the same time of day.

5. Results

Table 2 and Figure 1 compare actual announcements with matched non-announcement periods. Immediate seasonally adjusted trading activity, the logarithm of cumulative excess-volume area, and time to the seasonal baseline are higher after the actual release in all nine combinations of market and announcement type. All 27 paired differences are statistically significant at the 1% level. The largest differences occur following NFP announcements. Relative to the matched periods, immediate activity is higher by approximately 1.10 to 1.43 log points, and the seasonal volume baseline is reached approximately 30 to 34 minutes later. The corresponding time-to-baseline differences range from approximately 15 to 22 minutes for CPI and from 8 to 16 minutes for ISM.

Immediate activity also predicts subsequent excess volume during matched non-announcement periods. Short-horizon persistence in trading activity is therefore not unique to scheduled releases. The principal announcement effect is that the releases generate substantially larger initial activity shocks and larger cumulative responses than otherwise comparable periods at the same time of day. The slope comparisons provide some additional evidence for NFP. In DAX and CAC futures, the relation between immediate activity and subsequent excess volume is stronger following an actual NFP release than during its matched non-announcement period. The differences in the activity slopes are 0.190 for DAX ($p = 0.022$) and 0.273 for CAC ($p = 0.012$). The DAX CPI slope difference is positive but only weakly significant at the 10% level; no other slope difference is statistically significant.

The size and duration of the announcement-induced response also differ across release types. Table 3 and Figure 2 show that NFP produces the largest average response in every market. Mean immediate activity following NFP ranges

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Table 2
Actual Announcements and Matched Non-Announcement Periods

Market	Announcement	Matched pairs	Δ Activity	$\Delta \log(1 + \mathrm{AUC}^{120})$	Δ Time to baseline
Panel A: Differences in outcome levels					
FTSE	CPI	257	0.763***	0.339***	18.4***
FTSE	ISM	278	0.394***	0.320***	16.0***
FTSE	NFP	290	1.105***	0.530***	30.1***
DAX	CPI	257	0.852***	0.239***	14.8***
DAX	ISM	281	0.443***	0.295***	8.5***
DAX	NFP	290	1.157***	0.616***	33.4***
CAC	CPI	251	0.994***	0.494***	21.6***
CAC	ISM	236	0.694***	0.469***	15.5***
CAC	NFP	258	1.430***	0.590***	34.1***

Market	Announcement	Matched-period activity slope	Additional announcement slope	Interaction p
Panel B: Activity-propagation slope comparisons				
FTSE	CPI	0.419	0.060	0.383
FTSE	ISM	0.732	−0.082	0.476
FTSE	NFP	0.563	−0.013	0.866
DAX	CPI	0.351	0.100*	0.069
DAX	ISM	0.811	−0.197	0.139
DAX	NFP	0.504	0.190**	0.022
CAC	CPI	0.438	−0.087	0.148
CAC	ISM	0.619	−0.011	0.896
CAC	NFP	0.386	0.273**	0.012

Panel A reports actual-minus-matched-period mean differences using fixed intervals: the first five minutes for initial activity and the following 120 minutes for persistence. Time to baseline is measured in minutes. Panel B stacks actual and matched observations within each market and announcement type. The matched-period slope is the relation between immediate activity and subsequent persistence outside announcement periods. The additional slope is the change in this relation following an actual announcement. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$.

from 1.59 to 1.80, compared with 0.96 to 1.10 for CPI and 0.46 to 0.65 for ISM. NFP also produces the largest mean cumulative excess-volume area and the longest median time to the seasonal baseline in FTSE, DAX, and CAC futures. This ranking is consistent with previous evidence that NFP is among the most influential U.S. macroeconomic releases across financial markets (Andersen et al., 2003; Chaboud et al., 2008). The analysis does not attempt to identify why NFP dominates the other releases. Its broad information content, including payroll employment, unemployment, earnings, and revisions, nevertheless makes the result economically plausible.

The central regression result is that immediate trading activity strongly predicts the subsequent cumulative response. The separate estimates in Table 4 and Figure 3 are positive in all nine combinations and range from 0.285 to 0.474. A joint test rejects equality of the nine activity coefficients ($p < 0.001$). The activity relation is therefore common in direction but differs in magnitude across markets and announcement types.

The pooled estimates in Table 5 show the incremental contribution of activity. Announcement characteristics and pre-release Garman–Klass volatility explain 13.0% of the variation in the persistence outcome. Adding immediate activity raises adjusted R^2 from 0.130 to 0.395, an increase of 26.5 percentage points. The standardized activity coefficient is 0.350 ($p < 0.001$) before directional concentration is added and 0.357 ($p < 0.001$) after it is included.

Directional concentration provides virtually no additional explanatory power. Its pooled coefficient is 0.011 and statistically insignificant. Adding D raises adjusted R^2 from 0.3953 to 0.3959, an increase of approximately 0.06 percentage points. The separate estimates in Table 4 also lack a stable pattern. Only the FTSE CPI coefficient

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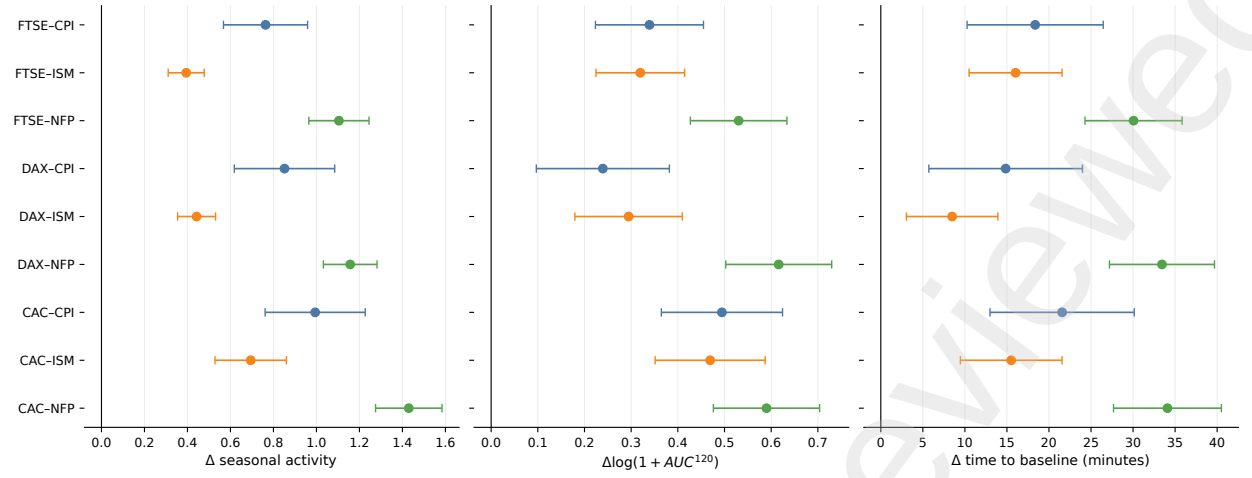


Figure 1: Announcement effects relative to matched non-announcement periods. The three panels show differences in immediate seasonally adjusted trading activity, $\log(1 + \mathrm{AUC}^{120})$, and time to the seasonal baseline. Each point represents one market and announcement type, and the lines show 95% confidence intervals based on HAC standard errors. All paired mean differences are statistically significant at the 1% level.

Table 3
Selected Descriptive Statistics

Market	Announcement	N	T^*	Mean activity	Mean AUC_{120}	Median TTB	Censored (%)
FTSE	CPI	257	2	0.960	57.48	33.0	15.6
FTSE	ISM	287	7	0.457	39.19	17.0	9.1
FTSE	NFP	297	3	1.588	61.51	54.0	15.2
DAX	CPI	257	2	1.099	54.49	28.0	12.1
DAX	ISM	281	2	0.551	37.40	16.0	8.9
DAX	NFP	298	3	1.716	64.63	56.5	14.1
CAC	CPI	253	13	1.091	64.37	34.0	15.8
CAC	ISM	257	4	0.651	45.52	19.0	9.7
CAC	NFP	274	6	1.796	70.86	56.5	18.2

The statistics use the primary 20-day seasonal benchmark, same-day contract activation, and estimated turning points. AUC_{120} is cumulative positive seasonally adjusted log volume over 120 consecutive one-minute observations beginning at T^*. TTB is time to the 125% seasonal baseline threshold and is right-censored at 120 minutes. Censored is the percentage of events that do not reach the threshold within the interval.

is significant before multiple-testing adjustment, and none of the nine directional coefficients remains significant after either Benjamini–Hochberg or Holm adjustment. The evidence therefore supports H2 but does not support the directional-flow prediction in H3. Persistence is more closely associated with the size of the initial participation shock than with the extent to which initial trading is concentrated on one side of the market.

Forecast dispersion provides an additional link between the pre-announcement information environment and the cumulative response. Table 4 shows that the primary NFP dispersion coefficients are −0.228 for FTSE, −0.149 for DAX, and −0.074 for CAC. The FTSE and DAX coefficients are significant at the 1% level, and the CAC coefficient is significant at the 5% level before adjustment. Under the Benjamini–Hochberg procedure, all three NFP coefficients remain significant at the 5% level. Under the more conservative Holm procedure, the FTSE and DAX coefficients remain significant, while the CAC coefficient does not. NFP therefore produces the largest average activity response, but its cumulative effect is weaker when professional forecasts are more dispersed.

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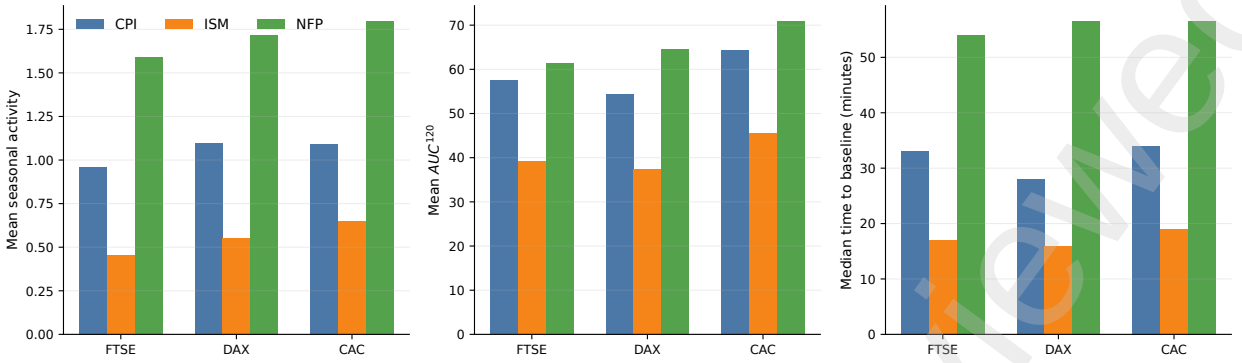


Figure 2: Seasonally adjusted trading activity following macroeconomic announcements. The panels show mean immediate activity, mean positive seasonal volume area during the 120-minute persistence interval, and median time to the 125% seasonal baseline threshold. The figure uses the primary same-day contract-activation rule and the turning point estimated for each market and announcement type.

Table 4
Separate Persistence Regressions by Market and Announcement Type

Market	Announcement	N	Activity	Directional concentration D	Forecast dispersion	Adjusted R^2
FTSE	CPI	257	0.313***	0.084**	0.037	0.221
			(0.042)	(0.039)	(0.025)	
FTSE	ISM	287	0.340***	−0.008	−0.007	0.274
			(0.052)	(0.033)	(0.033)	
FTSE	NFP	297	0.308***	0.061	−0.228***	0.349
			(0.037)	(0.039)	(0.079)	
DAX	CPI	257	0.357***	0.007	0.018	0.324
			(0.031)	(0.035)	(0.026)	
DAX	ISM	281	0.285***	−0.055	−0.038	0.194
			(0.043)	(0.047)	(0.046)	
DAX	NFP	297	0.416***	−0.007	−0.149***	0.434
			(0.041)	(0.021)	(0.025)	
CAC	CPI	253	0.322***	0.003	0.017	0.318
			(0.029)	(0.024)	(0.041)	
CAC	ISM	257	0.384***	−0.049	−0.021	0.321
			(0.034)	(0.053)	(0.036)	
CAC	NFP	274	0.474***	0.046	−0.074**	0.528
			(0.057)	(0.039)	(0.031)	

Each row reports a separate HAC regression of $\log(1 + \mathrm{AUC}^{120})$ on absolute surprise, forecast dispersion, pre-release Garman–Klass volatility, immediate activity, and directional concentration. Parentheses contain HAC standard errors. Stars refer to unadjusted p-values. None of the directional-concentration coefficients remains significant after Benjamini–Hochberg or Holm adjustment. All three NFP dispersion coefficients remain significant under Benjamini–Hochberg adjustment; the FTSE and DAX coefficients also remain significant under Holm adjustment. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$.

No comparable cross-market pattern appears for ISM or CPI. Forecast dispersion is largely unrelated to ISM persistence, while the CPI estimates do not share a common sign across the three markets. The pooled interaction between absolute surprise and forecast dispersion is also insignificant, indicating that the main NFP result is additive rather than driven by a systematic change in the effect of surprise magnitude. The negative NFP coefficients support the signal-precision prediction in H4b: greater pre-announcement dispersion is associated with a smaller cumulative excess-volume response after controlling for immediate activity and surprise magnitude. The differences-of-opinion prediction in H4a is not supported for NFP, and neither competing prediction describes all three announcement types.

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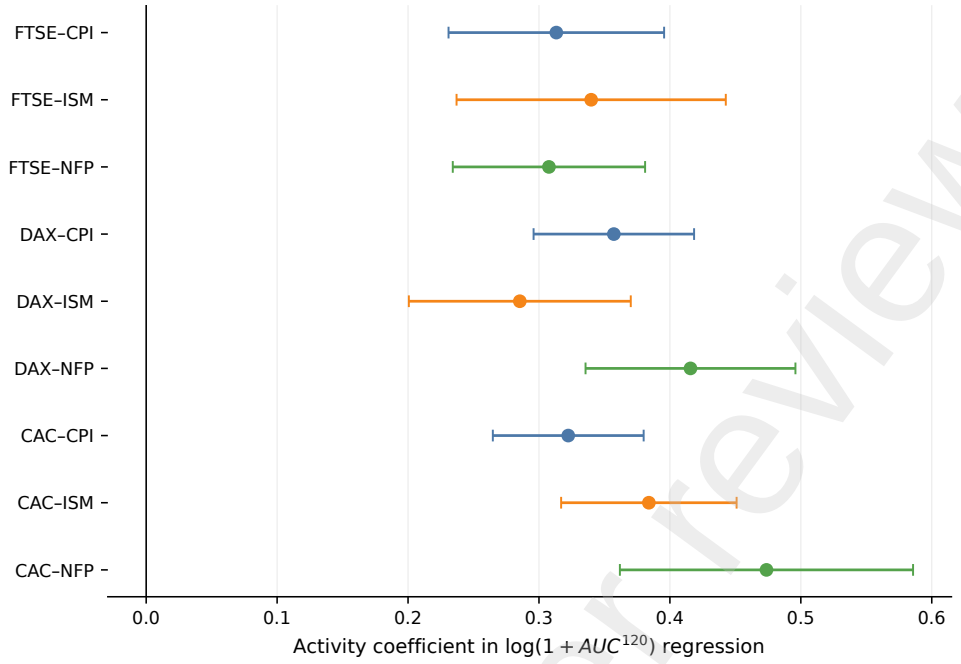


Figure 3: Separate activity coefficients for each market and announcement type. Each point is the coefficient on standardized immediate seasonally adjusted trading activity in a HAC regression of $\log(1 + \mathrm{AUC}^{120})$. The lines show 95% confidence intervals. Each model controls for absolute surprise, forecast dispersion, pre-release Garman–Klass volatility, and directional concentration D.

Table 5
Pooled Persistence Regressions

	P1 Controls	P2 Activity	P3 Activity+Direction	P4 VIX state
Absolute surprise	−0.013	−0.072***	−0.073***	−0.058**
Forecast dispersion	−0.056*	−0.021	−0.020	−0.013
Immediate activity	—	0.350***	0.357***	0.347***
Directional concentration D	—	—	0.011	0.013
Prior-day VIX	—	—	—	−0.056***
Activity × VIX	—	—	—	0.033
Market × announcement fixed effects	Yes	Yes	Yes	Yes
Standard errors clustered by announcement date	Yes	Yes	Yes	Yes
N	2,461	2,461	2,460	2,460
Adjusted R^2	0.130	0.395	0.396	0.402

The dependent variable is $\log(1 + \mathrm{AUC}^{120})$. All models include standardized pre-release Garman–Klass volatility. Activity, directional concentration, and Garman–Klass volatility are standardized within each market and announcement type. Surprise and forecast dispersion are standardized within announcement type. VIX is the standardized log closing level from the most recent U.S. trading day before the announcement. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$.

6. Robustness

The activity result is robust to the principal choices used to construct the continuous futures series, adjust volume seasonality, define the persistence interval, select the sample, and conduct inference. Activating the new futures contract on the trading day after the daily-volume crossover leaves the FTSE and DAX activity estimates nearly unchanged and preserves positive and statistically significant coefficients in CAC. The magnitude of the CAC CPI coefficient is more sensitive to the activation rule, but its sign and significance are not. Directional concentration, by contrast, has unstable

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signs and significance across the two activation rules, reinforcing the pooled evidence that it does not provide a general explanation of persistence.

Replacing the 20-day seasonal benchmark with 40 preceding non-announcement trading days leaves all nine activity coefficients positive and statistically significant. The same is true when the initial phase is fixed at the first five minutes after the release and persistence is measured over the following 120 minutes. The negative NFP dispersion coefficients also remain under both alternatives. A shorter persistence measure ending 60 minutes after the estimated turning point produces the same qualitative conclusions.

The activity coefficients remain positive and significant when 2020 is excluded, when contract-roll dates are removed, when observations with absolute surprises greater than three standard deviations are excluded, and when the CAC sample begins in 2005. The NFP dispersion result is especially stable in FTSE and DAX. The CAC estimate retains a negative sign but is more sensitive to excluding 2020, consistent with the wider uncertainty surrounding its coefficient.

The results are also insensitive to winsorization, year fixed effects, and alternative inference procedures. Winsorizing the continuous variables at the 1st and 99th percentiles, including year indicators, varying the HAC bandwidth from 2 to 8, or using HC3 standard errors does not alter the main conclusions. The activity result is therefore not attributable to a particular treatment of extreme observations or serial dependence.

Prior global uncertainty has a modest level effect but does not change activity propagation. The standardized log VIX from the most recent U.S. trading day before the announcement has a pooled coefficient of -0.056 ($p = 0.002$). Conditional on the initial activity shock, cumulative excess volume is therefore somewhat lower when prior VIX is high. The activity–VIX interaction is insignificant, however, indicating that the relation between immediate activity and persistence is stable across global uncertainty states.

Additional tests provide no evidence of systematic surprise-sign asymmetry. Neither the negative-surprise indicator nor its interaction with absolute surprise is robustly significant. The pooled interaction between surprise magnitude and forecast dispersion is also insignificant. When the corresponding market-level interaction tests are considered jointly, Benjamini–Hochberg and Holm adjustments eliminate the scattered individual results. These findings support the use of absolute surprise and the additive forecast-dispersion coefficient in the primary models.

Finally, moving-block-bootstrap quantile regressions show that the activity coefficient is positive at the 25th, 50th, 75th, and 90th percentiles of the conditional persistence distribution in every market and announcement type. The relation is therefore not confined to events with low, median, or unusually high persistence. The coefficient often becomes smaller toward the upper tail, suggesting that an additional increase in immediate activity contains less incremental information among events whose persistence is already high.

Taken together, these checks show that the activity and NFP dispersion results are not sensitive to the principal data-construction, sample, outcome, or inference choices.

7. Economic Interpretation and Practical Implications

The results distinguish the scale of the initial volume response from the cumulative trading activity that follows. An announcement may initiate repositioning, order execution, and inventory transfer that continue after the initial phase. Because these processes can generate offsetting directional pressures, sustained trading activity need not be accompanied by concentrated directional flow. The strong activity coefficient and weak directional-concentration results are therefore consistent with an activity propagation interpretation: the scale of initial trading, rather than its directional concentration, predicts the subsequent cumulative excess-volume response.

The placebo evidence places this interpretation in context. Immediate activity predicts subsequent excess volume during matched non-announcement periods as well as after announcements, indicating that short-horizon persistence in trading activity is not specific to public news. Announcements are instead distinguished by substantially higher initial activity, larger cumulative excess-volume areas, and longer times to the seasonal baseline. For NFP in DAX and CAC, the activity–persistence slope is also significantly steeper than in the corresponding matched periods, providing some evidence of stronger propagation in these two settings. Across the nine market–announcement combinations, however, the most consistent announcement effect is a shift in the level and duration of elevated activity rather than a uniformly steeper activity–persistence relation.

The forecast-dispersion results add an information-quality dimension to this interpretation of persistent activity. NFP produces the largest average response, yet greater disagreement among professional forecasters is associated with a smaller cumulative excess-volume area after controlling for immediate activity and surprise magnitude. This

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pattern is consistent with a less precise consensus benchmark reducing the weight placed on the released value. It is not consistent with the simple prediction that greater ex ante disagreement necessarily creates more persistent trading. The absence of a common ISM or CPI pattern also shows that forecast dispersion does not have an identical meaning across macroeconomic releases.

For execution desks and liquidity providers, immediate seasonally adjusted activity offers an observable summary of the trading environment that follows a release. It contains substantially more information about cumulative excess volume than directional concentration. The measure may therefore be useful for capacity allocation, participation rates, and inventory planning around scheduled announcements. The practical implication is deliberately limited: higher cumulative excess volume indicates that trading remains unusually active, not that execution costs must be higher.

The state differences across announcement types may also matter for exchange operations and market surveillance. NFP consistently produces the largest and longest activity response in all three futures markets. Capacity planning based solely on domestic European announcements would therefore overlook a recurring source of substantial cross-border trading activity.

Several limitations qualify the interpretation. First, the long historical sample does not contain a uniform quote or limit-order-book record. Directional concentration is inferred from transaction-price changes and cannot identify trader type, private information, liquidity provision, or inventory positions. Second, cumulative volume persistence is not a measure of liquidity resiliency. A market can sustain high trading volume while spreads and depth improve; connecting the persistence measure to execution costs requires complete order-book data.

Third, the estimated turning point summarizes the average path for each market and announcement type. It is intended as a common measurement boundary rather than as an event-specific structural break. This design preserves comparability and prevents an event's own volume path from determining its measurement window. The main conclusions are unchanged when the data-driven boundary is replaced by a fixed initial interval.

Finally, the analysis concerns U.S. macroeconomic announcements transmitted to three European equity-index futures markets. This setting permits comparison of simultaneous responses to a common information event, but the results need not generalize to domestic policy announcements, individual securities, less liquid contracts, or announcements released outside continuous trading hours.

8. Conclusion

This paper examines the trading activity that persists after scheduled U.S. macroeconomic announcements in FTSE, DAX, and CAC futures. The analysis removes predictable within-day seasonality, separates the initial activity increase from a 120-minute persistence interval, and compares actual releases with non-announcement periods at the same time of day.

The announcements generate higher immediate activity, larger cumulative excess-volume areas, and longer times to the seasonal baseline in every market and announcement type. NFP produces the largest and longest response in all three markets. Immediate seasonally adjusted trading activity strongly predicts the subsequent cumulative response: all nine separate activity coefficients are positive and significant, and adding activity raises pooled adjusted R^2 by 26.5 percentage points.

Directional concentration provides virtually no incremental explanatory power. Its pooled coefficient is insignificant, none of the separate coefficients survives multiple-testing adjustment, and the measure adds approximately 0.06 percentage points to adjusted R^2. Post-announcement volume persistence is therefore more closely associated with the scale of the initial participation shock than with one-sided trading pressure.

Forecast dispersion further conditions the NFP response. Greater pre-announcement NFP dispersion predicts a smaller cumulative excess-volume area in all three markets, with the strongest and most stable evidence in FTSE and DAX. The result is consistent with a signal-precision channel in which a less precise consensus benchmark weakens the cumulative response to the release. Comparable cross-market evidence does not appear for ISM or CPI.

Taken together, the findings characterize post-announcement volume persistence as a state of unusually high trading activity. Its magnitude is associated primarily with the initial trading activity and, for NFP, with the precision of the pre-release consensus. Future research using synchronized quotes, complete order books, or trader identifiers could link this persistence measure to liquidity provision, inventory transfer, and execution costs.

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Data Availability Statement

The raw futures transactions and announcement data are obtained from commercial sources and are subject to licensing restrictions; they cannot be redistributed. The analytical code and, subject to the applicable licenses, derived event-level data can be provided upon reasonable request.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used Google Gemini for language editing and Python code optimization. The author reviewed and edited the output as needed and take full responsibility for the content of the published article.

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